MTH603

Assignment # 01 Solution 2023

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Question:01

Question # 01:

Use Newton Raphson method to find the root of

$f(x) = x^3 - 2x + 2$ *with* $x_o = 0.2$

(Perform only three iterations)

Solution:

Here

$x^3 - 2x + 2 = 0$

let:

$f(x) = x^3 - 2x + 2$

$\frac{d}{dx}(x^3 - 2x + 2) = 3x^2 - 2$

$f'(x) = 3x^2 - 2$

$x_o = 0.2$

1st *iteration*:

$f(x_0) = f(0.2) = (0.2)^3 - 2(0.2) + 2 = 1.608$

$f'(x_o) = f'(x_o) = 3(0.2)^2 - 2 = -1.88$

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$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 0.2 - \frac{1.608}{(-1.88)} = 1.0553$$

2nd *Iteration*:

$$f(x_1) = f(1.0553) = (1.0553)^3 - 2(1.0553) + 2 = 1.0647$$

$$f'(x_1) = f'(1.0553) = 3(1.0553)^2 - 2 = 1.3411$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 1.0553 - \frac{1.0647}{1.3411}$$

$$x_2 = 0.2614$$

3rd *Iteration*:

$$f(x_2) = f(0.2614) = (0.2614)^3 - 2(0.2614) + 2 = 1.495$$

$$f'(x_2) = f(0.2614) = 3(0.2614)^2 - 2 = -1.7949$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$x_3 = 0.2614 - \frac{1.495}{(-1.7949)}$$

$$x_3 = 1.0943$$

Question # 02:

Solve the system of equation by Jacobi method starting with (0,0,0)

$$10x + y + z = 24$$
$$-x + 20y + z = 21$$
$$-x - 2y + 100z = 300$$

$$10x + y + z = 24$$
$$-x + 20y + z = 21$$
$$-x - 2y + 100z = 300$$

From the above equations :

$$x_{k+1} = \frac{1}{10}(24 - y_k - z_k)$$
$$y_{k+1} = \frac{1}{20}(21 + x_k - z_k)$$
$$z_{k+1} = \frac{1}{100}(300 + x_k + 2y_k)$$
$$Initial\ \ gauss\ (x, y, z) = (0, 0, 0)$$

1st iteration :

$$x_1 = \frac{1}{10}[24 - 0 - 0] = \frac{1}{10}[24] = 2.4$$
$$y_1 = \frac{1}{20}[21 + 0 - 0] = \frac{1}{20}[21] = 1.05$$
$$z_1 = \frac{1}{100}[300 + 0 + 2(0)] = \frac{1}{100}[300] = 3$$

1st iteration :

$$x_1 = \frac{1}{10}[24 - 0 - 0] = \frac{1}{10}[24] = 2.4$$
$$y_1 = \frac{1}{20}[21 + 0 - 0] = \frac{1}{20}[21] = 1.05$$
$$z_1 = \frac{1}{100}[300 + 0 + 2(0)] = \frac{1}{100}[300] = 3$$

2nd Iteration :

$$x_2 = \frac{1}{10}[21 - (1.05) - (3)] = \frac{1}{10}[19.95] = 1.995$$
$$y_2 = \frac{1}{20}[21 + (2.4) - (3)] = \frac{1}{20}[20.4] = 1.02$$
$$z_2 = \frac{1}{100}[300 + (2.4) + 2(1.05)] = \frac{1}{100}[304.5] = 3.045$$